

Model Compression for Very Short-Term Solar Irradiance Forecasting Using Infrared Sky Images

Ifran Rahman Nijhum^{1,2,†}, Arnav Jaitly^{2,†}, David Lillis², and Soumyabrata Dev^{1,2,3}

¹*The ADAPT Research Centre, Dublin, Ireland*

²*School of Computer Science, University College Dublin, Dublin, Ireland*

³*School of Computer Science and Statistics, Trinity College Dublin, Dublin, Ireland*

[†]*These authors contributed equally to this work.*

Abstract

Accurate short-term solar irradiance forecasting is essential for reliable photovoltaic power integration. While convolutional neural networks (CNNs) have demonstrated strong performance for sky-image-based forecasting, deploying deep learning models on resource-constrained edge devices remains challenging due to their computational and memory requirements. However, the impact of model compression on solar forecasting accuracy and deployment efficiency has not been systematically investigated. In this work, we present a comprehensive benchmark of structured and unstructured pruning for infrared sky image-based solar irradiance forecasting using infrared sky images from the Girasol dataset. We evaluate four widely used CNN architectures ResNet-18, EfficientNet-B0, MobileNetV2, and MobileNetV3-Small along with a lightweight custom baseline model (TinyCNN) across multiple pruning levels ranging from 20% to 80%. Models are assessed using forecasting accuracy, model size, parameter reduction, and inference latency. Experimental results show that MobileNetV2 achieves the best baseline forecasting performance with an RMSE of 37.11. The best structured pruning configuration achieved a 48.87% parameter reduction while maintaining an RMSE within 10% of its corresponding unpruned model. Unstructured pruning achieved sparsity levels of up to 59.80% and substantially reduced compressed checkpoint size, although it provided limited inference speed improvements on conventional hardware. Overall, structured pruning offers the most favorable trade-off between forecasting accuracy and deployment efficiency. These findings demonstrate that significant model compression can be achieved with minimal performance degradation, enabling practical edge deployment of CNN-based solar irradiance forecasting systems.

Keywords: Solar irradiance forecasting, infrared sky imaging, convolutional neural networks, model pruning, edge deployment

1 Introduction

The increasing adoption of photovoltaic energy systems has made accurate short-term solar irradiance forecasting essential for grid stability, energy management, and power dispatch [Barhmi et al., 2026, Alnajadah et al., 2025]. Since rapid cloud movement can cause large intra-hour fluctuations in solar power generation, forecasting Global Solar Irradiance (GSI) remains an important research problem [Nie et al., 2024]. Ground-based sky imaging systems are particularly effective for short-term irradiance forecasting because they directly capture local cloud dynamics at high temporal resolution [Diagne et al., 2013, Paletta et al., 2023]. The Girasol dataset introduced by Terrén-Serrano et al. [Terrén-Serrano et al., 2021] provides synchronized infrared sky images and pyranometer measurements for irradiance estimation. Infrared imaging is useful in this context because it captures cloud structure while being less affected by strong solar glare than visible-light imaging.

Recent deep learning approaches have improved image-based irradiance forecasting by learning cloud-related features directly from sky images [Klotz and Nayar, 2026]. CNN-based models, including lightweight

architectures such as MobileNetV2, have shown promising performance on the Girasol dataset [Nijhum et al., 2024]. Complementary research has also leveraged RGB and near-infrared (NIR) sky images for solar forecasting, highlighting the value of multi-spectral imaging for capturing cloud dynamics and irradiance variability [Nijhum et al., 2026]. However, practical deployment remains challenging. Solar monitoring stations and embedded IoT devices often operate under limited memory, computation, and energy budgets, while many CNN-based forecasting models are still too costly for efficient edge inference [Menghani, 2023, Paletta et al., 2023]. This creates a trade-off between forecasting accuracy and deployment efficiency. Model compression provides a practical way to reduce this cost. Pruning can remove redundant parameters or channels from neural networks, while quantization reduces numerical precision to lower storage and computational requirements [He and Xiao, 2023, Fang et al., 2023, Wu et al., 2020, Guo, 2018]. Although these techniques are widely studied in general computer vision, their use for sky-image-based solar irradiance forecasting remains limited.

To the best of our knowledge, no prior work has systematically evaluated pruning techniques across multiple CNN architectures for infrared sky image-based solar irradiance forecasting. Existing studies mainly focus on forecasting accuracy, with limited attention to deployment-oriented metrics such as model size and inference latency. In this work¹, we present a systematic compression benchmark for CNN-based solar irradiance forecasting using infrared sky imagery from the Girasol dataset. Five architectures, ResNet18, MobileNetV2, MobileNetV3-Small, EfficientNet-B0, and a custom baseline (Tiny CNN), are evaluated under multiple structured and unstructured pruning levels. The models are compared using validation RMSE, model size, and inference latency.

The main contributions of this work are summarized as follows:

- We evaluate the performance of several popular lightweight CNN architectures for short-term solar irradiance forecasting using infrared sky imagery.
- We perform both structured and unstructured pruning across multiple compression levels to analyze their impact on forecasting performance.
- We provide a comparative evaluation of forecasting accuracy, model size, and inference latency to study the trade-off between compression and deployment efficiency.

2 Methodology

This study investigates the impact of structured and unstructured pruning on convolutional neural networks (CNNs) for infrared sky image-based solar irradiance forecasting. The overall pipeline consists of four stages: dataset preparation, baseline model training, model compression through pruning, and post-pruning fine-tuning and evaluation. Infrared sky images are used as model inputs, while future global horizontal irradiance (GHI) values measured by a pyranometer serve as prediction targets. Multiple lightweight CNN architectures are evaluated, including MobileNetV2, MobileNetV3-small, EfficientNet-B0, ResNet-18, and a custom Tiny CNN baseline. Baseline models are first trained using a unified training configuration. Structured and unstructured pruning are then independently applied at multiple pruning ratios, followed by fine-tuning and evaluation.

2.1 Problem Formulation

The forecasting task is formulated as a supervised image-to-regression problem. Given an infrared sky image captured at time step t , the objective is to predict the future solar irradiance value at time step $t + L$, where L denotes the forecasting horizon.

$$\hat{y}_{t+L} = f(I_t; \theta), \quad (1)$$

¹<https://github.com/ifran-rahman/solar-forecasting-pruning>

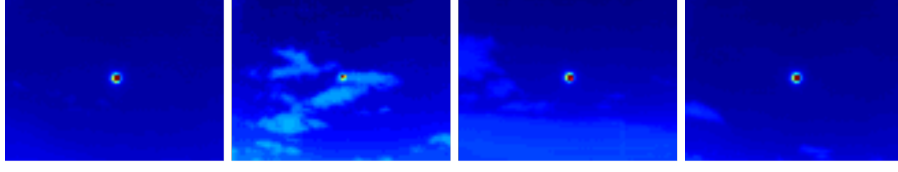


Figure 1: Sample infrared sky images from the Girasol dataset used for solar irradiance forecasting. The images capture a range of sky conditions, including clear, partially cloudy, and overcast scenarios.

where I_t denotes the infrared sky image at time t , $\hat{y}_{t+L}$ denotes the predicted future irradiance value, $f(\cdot)$ represents the CNN forecasting model, and θ denotes the trainable model parameters. The forecasting horizon was fixed at 5 minutes (20 future time steps).

2.2 Dataset Preparation

The Girasol Sky Imaging and Global Solar Irradiance Dataset [Terrén-Serrano et al., 2021] contains synchronized infrared sky images and pyranometer-based irradiance measurements collected at the University of New Mexico for short-term solar forecasting. Figure 1 demonstrate some sample IR images from the dataset. This study uses 59,778 infrared sky images acquired between December 2017 and January 2018. To preserve temporal consistency and prevent data leakage, the dataset was split chronologically, with training samples drawn from 8 December 2017 to 8 January 2018 and validation samples taken from the subsequent 7-day period, consisting of 48,164 raw training images and 11,614 raw validation images. The forecasting horizon was fixed at $L = 20$ future time steps. After removing samples without valid future targets, the final dataset contained 47,564 training samples and 11,474 validation samples. For each sample, the input image at time t was paired with the irradiance measurement at time $t + L$. Images with a resolution of 60×80 pixels were normalized using min-max normalization.

2.3 CNN Architectures

Five CNN architectures were evaluated to analyze the effectiveness of pruning across lightweight and heavy-weight models. These widely adopted architectures are commonly used benchmarks for studying accuracy-efficiency trade-offs. All architectures were adapted for scalar irradiance regression by replacing their original classification heads with a single fully connected regression layer producing one irradiance prediction.

MobileNetV2 is a lightweight CNN architecture based on inverted residual blocks and depthwise separable convolutions, designed for efficient mobile and edge inference. MobileNetV3 Small extends this design by incorporating squeeze-and-excitation modules and hardware-aware neural architecture search for improved computational efficiency. EfficientNet-B0 employs compound scaling of network depth, width, and input resolution to achieve a balanced trade-off between accuracy and computational cost. ResNet18 is a residual CNN architecture that uses skip connections to facilitate stable optimization of deeper networks and serves as the heavyweight reference model in this study. In addition, a custom lightweight architecture, referred to as TinyCNN, was implemented using two convolutional layers with ReLU activations and max-pooling operations, followed by a fully connected regression head. The TinyCNN architecture consists of two convolutional blocks followed by a fully connected regression layer, as summarized in Eq. (2). TinyCNN was designed as an extremely lightweight edge-deployment baseline.

$$\text{Input} \rightarrow \text{Conv}(3 \times 3, 16) \rightarrow \text{ReLU} \rightarrow \text{MaxPool} \rightarrow \text{Conv}(3 \times 3, 32) \rightarrow \text{ReLU} \rightarrow \text{MaxPool} \rightarrow \text{Flatten} \rightarrow \text{FC}(9600 \rightarrow 1) \quad (2)$$

All models were trained from scratch without pretrained weights. For MobileNetV2, MobileNetV3-Small, EfficientNet-B0, and ResNet18, the original classification layers were removed and replaced with regression heads. Adaptive average pooling was additionally applied after feature extraction for the MobileNet-based architectures.

2.4 Structured Pruning

Structured pruning removes entire convolutional filters or channels, reducing parameter count and computational complexity while preserving dense tensor operations compatible with standard hardware [Zhu et al., 2025]. Magnitude-based structured pruning was implemented using the `torch-pruning` framework with L2-norm filter importance:

$$S_i = \|W_i\|_2 = \sqrt{\sum_j w_{ij}^2}, \quad (3)$$

where W_i denotes the weights of filter i . Filters with lower importance scores were pruned first.

Pruning ratios of 20%, 40%, 60%, and 80% were evaluated across all CNN architectures. Following pruning, models were fine-tuned for 3 epochs using Adam with a reduced learning rate of 10^{-4} . Performance was evaluated using RMSE, MAE, parameter reduction, model size, and inference latency.

$$\text{Parameter Reduction (\%)} = \left(1 - \frac{P_{\text{pruned}}}{P_{\text{baseline}}}\right) \times 100, \quad (4)$$

where P_{baseline} and P_{pruned} denote the number of parameters before and after pruning, respectively.

2.5 Unstructured Pruning

Unstructured pruning removes individual weights while preserving the original tensor dimensions, producing sparse weight matrices. L1 unstructured pruning was applied to convolutional and linear layers using the PyTorch pruning utilities. Weight importance was determined using absolute magnitude:

$$S_{ij} = |w_{ij}|, \quad (5)$$

where w_{ij} denotes an individual weight parameter.

Pruning ratios of 20%, 40%, 60%, and 80% were evaluated. After pruning, models were fine-tuned for 3 epochs using Adam with a learning rate of 10^{-4} . Final evaluation included RMSE, MAE, sparsity percentage, model size, and inference latency.

2.6 Training Configuration

All models were trained using the Adam optimizer with a batch size of 256 and a mean squared error (MSE) loss function. To ensure reproducibility, a fixed random seed of 42 was used for NumPy, Python, and PyTorch, with deterministic CUDA operations enabled. The baseline models were trained from random initialization for 50 epochs using an initial learning rate of 0.001. A StepLR scheduler was employed to reduce the learning rate by a factor of 0.5 every 10 epochs, and the checkpoint corresponding to the lowest validation loss was retained for evaluation. For both structured and unstructured pruning experiments, the trained baseline models served as the starting point. Pruning was applied at four pruning levels (20%, 40%, 60%, and 80%), followed by three epochs of fine-tuning using a reduced learning rate of (1×10^{-4}) to recover performance. Structured pruning was implemented using the Torch-Pruning MagnitudePruner with L2-norm importance, while unstructured pruning employed L1-based weight pruning on convolutional and linear layers. In all cases, the best fine-tuned model was selected based on validation loss and used for the final evaluation.

3 Experiments and Results

3.1 Evaluation Metrics

All models were optimized using mean squared error (MSE) loss and evaluated using root mean squared error (RMSE) and mean absolute error (MAE), reported in W/m²:

Table 1: Baseline forecasting performance, parameter counts, and checkpoint sizes for 60×80 infrared sky images. Lower values indicate better forecasting performance.

Model	Params (M) ↓	RMSE ↓	MAE ↓	MSE ↓	Size (MB) ↓
TinyCNN	0.015	82.14	61.87	6747.77	0.06
MobileNetV2	2.225	37.11	27.14	1376.78	8.72
MobileNetV3	0.928	39.72	28.22	1577.53	3.67
EfficientNet-B0	4.009	38.97	25.20	1518.69	15.58
ResNet-18	11.177	45.49	33.76	2068.93	42.71

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (6)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (7)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|. \quad (8)$$

Here y_i and $\hat{y}_i$ denote the ground-truth and predicted irradiance values, respectively. In addition to forecasting accuracy, deployment-oriented metrics including parameter reduction, sparsity percentage, checkpoint size, compressed checkpoint size, and inference latency per image are also measured to evaluate model efficiency after compression.

3.2 Baseline Performance

The forecasting performance of the evaluated CNN architectures is presented in Table 1. All models were trained using infrared sky images with a spatial resolution of 60×80 pixels. MobileNetV2 achieved the lowest RMSE of 37.11 and the lowest MSE of 1376.78, indicating the strongest overall forecasting performance among the evaluated models. EfficientNet-B0 obtained the lowest MAE of 25.20, while its RMSE remained close to MobileNetV2. MobileNetV3 also achieved competitive performance, with an RMSE of 39.72 and a smaller checkpoint size of 3.67 MB. ResNet-18, despite having the largest checkpoint size of 42.71 MB, did not provide the best forecasting accuracy. TinyCNN required only 0.06 MB of storage, making it the most compact model, but its prediction error was noticeably higher than the other architectures. Overall, these results show that larger model size does not necessarily lead to better forecasting performance for low-resolution infrared sky images. MobileNetV2 provides the best accuracy, while MobileNetV3 offers a strong balance between accuracy and storage efficiency.

3.3 Structured Pruning

Structured pruning was applied to all baseline models using the torch-pruning framework at pruning levels of 20%, 40%, 60%, and 80%. Following pruning, each model was fine-tuned for three epochs and evaluated using RMSE, parameter reduction, model size, and inference latency. For each architecture, the best configuration was selected as the most compressed model that maintained an RMSE increase below 10% relative to the baseline model. For structured pruning, compression was quantified using parameter reduction. Table 2 and Figure 2a show that structured pruning can substantially reduce model complexity while largely preserving forecasting performance. The selected configurations achieved parameter reductions of approximately 35-49%, resulting in smaller model sizes with only modest accuracy degradation. TinyCNN achieved the highest parameter reduction (48.87%) at a 40% pruning level, with an RMSE increase of less than 5%. Among the larger architectures, ResNet-18 demonstrated the strongest robustness to pruning, maintaining near-baseline accuracy

Table 2: Best structured pruning configurations selected using an experimental RMSE increase threshold of 10%

Model	Prune (%)	Param. Red. (%)	Base RMSE	Final RMSE	Size (MB)	Latency (ms/img)
EfficientNet	20	35.37	38.97	40.97	10.15	0.0357
MobileNetV2	20	35.46	37.10	42.62	5.69	0.0232
MobileNetV3	20	35.19	39.72	48.52	2.42	0.0230
TinyCNN	40	48.87	82.14	86.12	0.03	0.0024
ResNet-18	20	36.26	45.48	46.40	27.25	0.0136

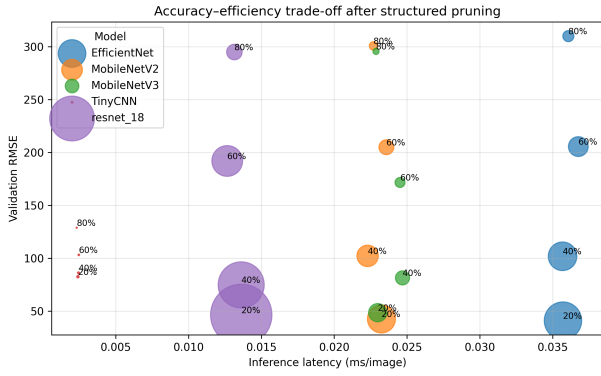
Table 3: Best unstructured pruning configurations selected using a maximum allowable RMSE increase of 10%.

Model	Prune (%)	Sparsity (%)	Base RMSE	Final RMSE	Size (MB)	Latency (ms/img)
EfficientNet	20	19.74	38.97	40.37	12.41	0.0371
MobileNetV2	20	19.69	37.10	38.23	6.94	0.0233
MobileNetV3	20	19.68	39.72	52.30	2.90	0.0225
TinyCNN	60	59.80	82.14	83.50	0.03	0.0025
ResNet-18	20	19.98	45.48	49.02	34.18	0.0120

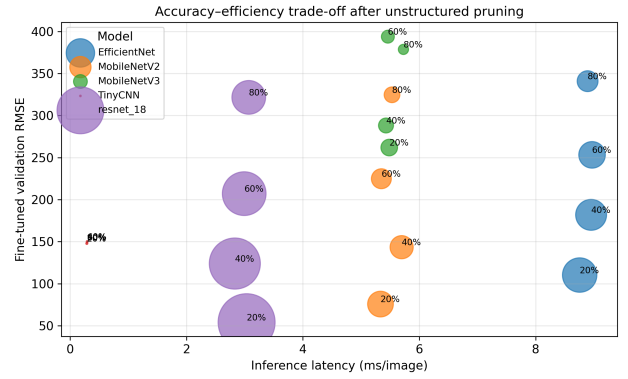
despite a 36.26% reduction in parameters. As the pruning level increased from 20% to 80%, validation RMSE generally increased across all models, indicating a trade-off between compression and forecasting accuracy. Moderate pruning levels (20-40%) provided the best balance between efficiency and performance, while inference latency changed only marginally, suggesting that parameter reduction does not necessarily translate into proportional speed improvements.

3.4 Unstructured Pruning

To evaluate the compressibility of the forecasting models, unstructured pruning was applied using L1-norm weight pruning on all convolutional and fully connected layers. Each model was pruned at 20%, 40%, 60%, and 80% sparsity levels and subsequently fine-tuned for 3 epochs. Since unstructured pruning preserves the original network architecture and does not reduce the number of stored parameters, compression effectiveness was evaluated using the size reduction of gzip-compressed model checkpoints. Structured pruning reports the actual checkpoint size because parameters are physically removed from the network architecture. In contrast, unstructured pruning reports gzip-compressed checkpoint size, as sparsity alone does not reduce storage requirements when the original dense parameter tensors are retained. For each architecture, the best configuration was selected as the most compressed model that maintained an RMSE increase below 10% relative to the baseline model. For unstructured pruning, compression was quantified using model sparsity. Table 3 summarizes the selected configurations. TinyCNN exhibited the highest pruning tolerance, achieving 59.80% sparsity while slightly improving forecasting accuracy. MobileNetV3 also benefited from pruning, achieving an 18.68% improvement in RMSE at 39.35% sparsity. EfficientNet achieved 39.49% sparsity with only a 9.65% RMSE increase while reducing the compressed checkpoint size by 30.35%. In contrast, MobileNetV2 and ResNet-18 were more sensitive to pruning and could not satisfy the 10% RMSE constraint at any tested pruning level. Figure 2b illustrates the resulting accuracy–efficiency trade-off. Increasing sparsity generally improved checkpoint compression but produced only minor changes in inference latency because the network architecture remained unchanged. These results show that unstructured pruning can substantially reduce model storage requirements while preserving forecasting performance at moderate sparsity levels, but it provides limited inference speed benefits on standard dense hardware.



(a) Accuracy–efficiency trade-off after structured pruning. Bubble size is proportional to model size.



(b) Accuracy–efficiency trade-off after unstructured pruning. Bubble size is proportional to the compressed checkpoint size.

Figure 2: Accuracy–efficiency trade-offs obtained using structured and unstructured pruning. Lower RMSE and lower inference latency indicate better forecasting efficiency, while bubble size represents the storage footprint of each pruned model.

4 Discussion and Conclusion

The experimental results demonstrate that effective solar irradiance forecasting from low-resolution infrared sky images can be achieved using relatively lightweight CNN architectures. Among the baseline models, MobileNetV2 provided the best overall forecasting accuracy, while MobileNetV3 achieved comparable performance with a smaller storage footprint. In contrast, the substantially larger ResNet-18 model did not deliver superior forecasting accuracy, suggesting that increasing model complexity does not necessarily improve performance for this task. The pruning experiments further revealed that considerable redundancy exists within the evaluated networks. Structured pruning successfully reduced parameter counts and model sizes while maintaining acceptable forecasting accuracy, with moderate pruning levels generally providing the best balance between compression and performance. TinyCNN exhibited the highest pruning tolerance, whereas larger architectures such as ResNet-18 remained surprisingly robust despite substantial parameter removal. A comparison of the two pruning strategies highlights important deployment trade-offs. Structured pruning reduced model complexity by removing channels and filters, leading to smaller models and modest improvements in inference efficiency. In contrast, unstructured pruning primarily reduced compressed checkpoint size through weight sparsity while producing little change in inference latency because the original network architecture remained unchanged. These findings indicate that structured pruning is more suitable for deployment on conventional edge devices, whereas unstructured pruning is most beneficial when storage reduction is the primary objective or when sparse-computation hardware is available. Overall, the results show that substantial model compression can be achieved with only limited degradation in forecasting accuracy, making lightweight CNNs combined with pruning a practical solution for resource-constrained solar forecasting systems. Future work will investigate knowledge distillation, quantization, and multimodal forecasting approaches to further improve both efficiency and predictive performance.

Acknowledgments

This research was conducted with the financial support of Research Ireland under Grant Agreement No. 13/RC/2106_P2 at the ADAPT Centre at Trinity College Dublin. ADAPT, the Research Ireland Centre for AI-Driven Digital Content Technology, is funded through the Research Ireland Centres Programme.

References

- [Alnajadah et al., 2025] Alnajadah, A., Bremer, C., Butler, S., Li, F., and TamizhMani, G. (2025). Intra-hour solar irradiance forecasting using sky cameras for pv plants. *Sustainable Energy Technologies and Assessments*, 84:104743.
- [Barhmi et al., 2026] Barhmi, K., Golroodbari, S. M., Knap, W., and Van Sark, W. (2026). Real-time solar irradiance forecasting for grid integration using all-sky imagery and multi-stage ai with kalman filter optimization. *Renewable Energy*, 259:125117.
- [Diagne et al., 2013] Diagne, M., David, M., Lauret, P., Boland, J., and Schmutz, N. (2013). Review of solar irradiance forecasting methods and a proposition for small-scale insular grids. *Renewable and Sustainable Energy Reviews*, 27:65–76.
- [Fang et al., 2023] Fang, G., Ma, X., Song, M., Mi, M. B., and Wang, X. (2023). Depgraph: Towards any structural pruning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 16091–16101.
- [Guo, 2018] Guo, Y. (2018). A survey on methods and theories of quantized neural networks. *arXiv preprint arXiv:1808.04752*.
- [He and Xiao, 2023] He, Y. and Xiao, L. (2023). Structured pruning for deep convolutional neural networks: A survey. *IEEE transactions on pattern analysis and machine intelligence*, 46(5):2900–2919.
- [Klotz and Nayar, 2026] Klotz, J. and Nayar, S. K. (2026). Forecasting solar energy using a single image. *Solar Energy*, 313:114540.
- [Menghani, 2023] Menghani, G. (2023). Efficient deep learning: A survey on making deep learning models smaller, faster, and better. *ACM Computing Surveys*, 55(12):1–37.
- [Nie et al., 2024] Nie, Y., Li, X., Paletta, Q., Aragon, M., Scott, A., and Brandt, A. (2024). Open-source sky image datasets for solar forecasting with deep learning: A comprehensive survey. *Renewable and Sustainable Energy Reviews*, 189:113977.
- [Nijhum et al., 2024] Nijhum, I. R., Kaloni, D., Kenny, P., and Dev, S. (2024). Enhancing intra-hour solar irradiance estimation through knowledge distillation and infrared sky images. In *IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium*, pages 7207–7211.
- [Nijhum et al., 2026] Nijhum, I. R., Scanlon, C., and Dev, S. (2026). Design and evaluation of a dual-spectrum low-cost sky imager for enhanced solar prediction. *Measurement: Energy*, page 100103.
- [Paletta et al., 2023] Paletta, Q., Terrén-Serrano, G., Nie, Y., Li, B., Bieker, J., Zhang, W., Dubus, L., Dev, S., and Feng, C. (2023). Advances in solar forecasting: Computer vision with deep learning. *Advances in Applied Energy*, 11:100150.
- [Terrén-Serrano et al., 2021] Terrén-Serrano, G., Bashir, A., Estrada, T., and Martínez-Ramón, M. (2021). Girasol, a sky imaging and global solar irradiance dataset. *Data in Brief*, 35:106914.
- [Wu et al., 2020] Wu, H., Judd, P., Zhang, X., Isaev, M., and Micikevicius, P. (2020). Integer quantization for deep learning inference: Principles and empirical evaluation. *arXiv preprint arXiv:2004.09602*.
- [Zhu et al., 2025] Zhu, K., Hu, F., Ding, Y., Zhou, W., and Wang, R. (2025). A comprehensive review of network pruning based on pruning granularity and pruning time perspectives. *Neurocomputing*, 626:129382.